

Topic 9: Hypothesis Testing: Two-sample Tests

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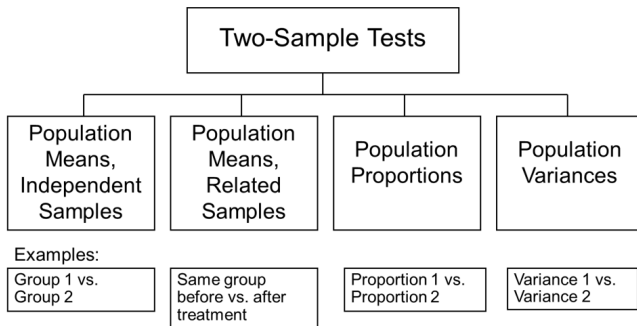
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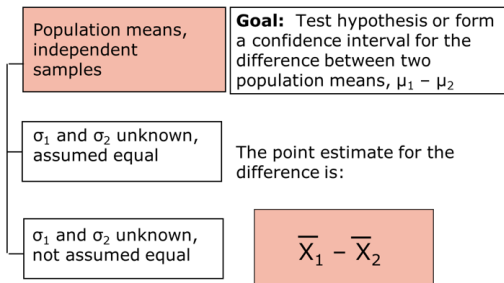


Two-sample Tests



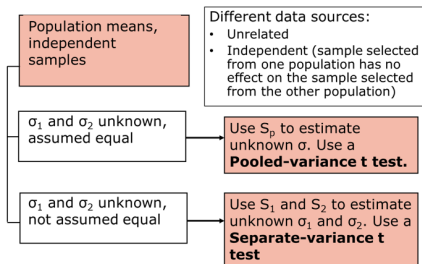
Difference Between Two Means

- The **Z test for the difference between two means** is a test statistic used in hypothesis tests about the difference between means of two populations.



Difference Between Two Means: Independent Samples

- Different data sources
 - Unrelated
 - Independent
 - ▷ Sample selected from one population has no effect on the sample selected from the other population.



Hypothesis Tests for Two Population Means (1 of 2)

Two Population Means, Independent Samples

Lower-tail test:

$$H_0 : \mu_1 \geq \mu_2$$

$$H_1 : \mu_1 < \mu_2$$

i.e.,

$$H_0 : \mu_1 - \mu_2 \geq 0$$

$$H_1 : \mu_1 - \mu_2 < 0$$

Upper-tail test:

$$H_0 : \mu_1 \leq \mu_2$$

$$H_1 : \mu_1 > \mu_2$$

i.e.,

$$H_0 : \mu_1 - \mu_2 \leq 0$$

$$H_1 : \mu_1 - \mu_2 > 0$$

Two-tail test:

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

i.e.,

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_1 : \mu_1 - \mu_2 \neq 0$$



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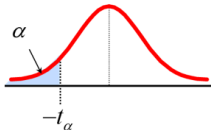
Hypothesis Tests for Two Population Means (2 of 2)

Two Population Means, Independent Samples

Lower-tail test:

$$H_0 : \mu_1 - \mu_2 \geq 0$$

$$H_1 : \mu_1 - \mu_2 < 0$$

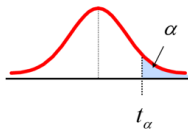


Reject H_0 if $t_{STAT} < -t_\alpha$

Upper-tail test:

$$H_0 : \mu_1 - \mu_2 \leq 0$$

$$H_1 : \mu_1 - \mu_2 > 0$$

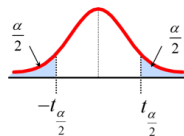


Reject H_0 if $t_{STAT} > t_\alpha$

Two-tail test:

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_1 : \mu_1 - \mu_2 \neq 0$$



Reject H_0 if $t_{STAT} < -t_{\frac{\alpha}{2}}$

OR $t_{STAT} > t_{\frac{\alpha}{2}}$



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Hypothesis Tests for Two Population Means (assume $\sigma_1 = \sigma_2$) (1 of 3)

Assumptions:

- Samples are randomly and independently drawn.
- Populations are normally distributed or both sample sizes are at least 30.
- Population variances are unknown but assumed equal.



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Hypothesis Tests for Two Population Means (assume $\sigma_1 = \sigma_2$) (2 of 3)

- The pooled variance is:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)}$$

- The test statistic is:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where

$$t \text{ has d.f.} = (n_1 + n_2 - 2)$$



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Hypothesis Tests for Two Population Means (assume $\sigma_1 = \sigma_2$) (3 of 3)

- The confidence interval for $\mu_1 - \mu_2$ is:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\frac{\alpha}{2}} \sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

where

$$t_{\frac{\alpha}{2}} \text{ has d.f. } = (n_1 + n_2 - 2)$$



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Example: Pooled-Variance t Test

You are a financial analyst for a brokerage firm. Is there a difference in dividend yield between stocks listed on the NYSE & NASDAQ? You collect the following data:

	NYSE	NASDAQ
Number	21	25
Sample mean	3.27	2.53
Sample std dev	1.30	1.16

Assuming both populations are approximately normal with equal variances, is there a difference in mean yield ($\alpha = 0.05$)?



Solution (1 of 3)

- $H_0 : \mu_1 - \mu_2 = 0$ i.e. $(\mu_1 = \mu_2)$
 $H_1 : \mu_1 - \mu_2 \neq 0$ i.e. $(\mu_1 \neq \mu_2)$
- Pooled variance is:

$$\begin{aligned} S_p^2 &= \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)} \\ &= \frac{(21 - 1)1.30^2 + (25 - 1)1.16^2}{(21 - 1) + (25 - 1)} = 1.5021 \end{aligned}$$

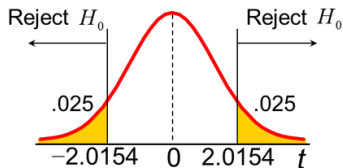
- The test statistic is:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2(\frac{1}{n_1} + \frac{1}{n_2})}} = \frac{(3.27 - 2.53) - 0}{\sqrt{1.5021(\frac{1}{21} + \frac{1}{25})}} = \boxed{2.040}$$



Solution (2 of 3)

- $\alpha = 0.05$, $df = 21 + 25 - 2 = 44$
- Critical values: $t = \pm 2.0154$
- Decision: Reject H_0 at $\alpha = 0.05$
- Conclusion: There is evidence of a difference in means.



Solution (3 of 3)

- Since we rejected H_0 can we be 95% confident that $\mu_{NYSE} > \mu_{NASDAQ}$?
- 95% confidence interval for $\mu_{NYSE} - \mu_{NASDAQ}$:

$$\begin{aligned}(\bar{X}_1 - \bar{X}_2) \pm t_{\frac{\alpha}{2}} \sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} &= 0.74 \pm 2.0154 \times 0.3628 \\ &= (0.009, 1.471)\end{aligned}$$

- Since 0 is less than the entire interval, we can be 95% confident that $\mu_{NYSE} > \mu_{NASDAQ}$



Hypothesis Tests for Two Population Means (assume $\sigma_1 \neq \sigma_2$) (1 of 3)

Assumptions:

- Samples are randomly and independently drawn.
- Populations are normally distributed or both sample sizes are at least 30.
- Population variances are unknown and cannot be assumed to be equal.



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Hypothesis Tests for Two Population Means (assume $\sigma_1 \neq \sigma_2$) (2 of 3)

- This test utilises two separate sample variances to estimate the degrees of freedom for the t test.
- The test statistic is:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

where

$\bar{X}_1$ = mean of the sample taken from population 1

S_1^2 = variance of the sample taken from population 1

n_1 = size of the sample taken from population 1

$\bar{X}_2$ = mean of the sample taken from population 2

S_2^2 = variance of the sample taken from population 2

n_2 = size of the sample taken from population 2



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Hypothesis Tests for Two Population Means (assume $\sigma_1 \neq \sigma_2$) (3 of 3)

The separate variance t test statistic follows

- approximately a t distribution
- with degrees of freedom, ν , equal to the *integer* portion of the following calculation

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2-1}}$$



Example: Separate Variance t Test

You are a financial analyst for a brokerage firm. Is there a difference in dividend yield between stocks listed on the NYSE & NASDAQ? You collect the following data:

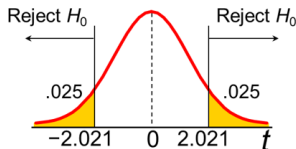
	NYSE	NASDAQ
Number	21	25
Sample mean	3.27	2.53
Sample std dev	1.30	1.16

Assuming both populations are approximately normal with unequal variances, is there a difference in mean yield ($\alpha = 0.05$)?



Solution

- $H_0 : \mu_1 - \mu_2 = 0$ i.e. $(\mu_1 = \mu_2)$
 $H_1 : \mu_1 - \mu_2 \neq 0$ i.e. $(\mu_1 \neq \mu_2)$
- $\alpha = 0.05$, $df = 40$
- t test statistic: 2.019
- Critical values: $t = \pm 2.021$
- Decision: Fail to reject H_0 at $\alpha = 0.05$
- Conclusion: There is insufficient evidence of a difference in means.



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Excel Output

	A	B	C	D	E	F	G	H	I	J
1	Separate-Variances t Test									
2	(assumes unequal population variances)									
3	Data									
4	Hypothesized Difference	0								
5	Level of Significance	0.05								
6	Population 1 Sample									
7	Sample Size	21	=COUNT(DATACOPY(\$A:\$A))							
8	Sample Mean	3.27	=AVERAGE(DATACOPY(\$A:\$A))							
9	Sample Standard Deviation	1.3	=STDEV.S(DATACOPY(\$A:\$A))							
10	Population 2 Sample									
11	Sample Size	25	=COUNT(DATACOPY(\$B:\$B))							
12	Sample Mean	2.53	=AVERAGE(DATACOPY(\$B:\$B))							
13	Sample Standard Deviation	1.16	=STDEV.S(DATACOPY(\$C:\$C))							
14	Intermediate Calculations									
15	Pop. 1 Sample Variance	1.69	=B9^2							
16	Pop. 2 Sample Variance	1.3456	=B13^2							
17	Pop. 1 Sample Variance / Sample Size	0.0804762	=B16/B7							
18	Pop. 2 Sample Variance / Sample Size	0.053824	=B17/B11							
19	Numerator of Degrees of Freedom	0.0180365	=B18+B19^2							
20	Denominator of Degrees of Freedom	0.0004445	=B18^2/(B7-1) + (B19^2)/(B11-1)							
21	Total Degrees of Freedom	40.574333	=B20/B21							
22	Degrees of Freedom	40	=INT(B22)							
23	Separate Variance Denominator	0.3665	=SQRT(B18+B19)							
24	Difference in Sample Means	0.74	=B8 - B12							
25	t Test Statistic	2.019	=B25 - B4)/B24							
26	Two-Tail Test									
27	Lower Critical Value	-2.021	=-T.INV.2T(B5,B23)							
28	Upper Critical Value	2.021	=T.INV.2T(B5,B23)							
29	p-value	0.050	=T.DIST.2T(ABS(B26),B23)-B4							
30	Do not reject the null hypothesis									
31	=IF(B31<B5, "Reject the null hypothesis", "Do not reject the null hypothesis")									



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You Do It

The financial aid office of a university asks a sample of students about their employment and earnings. The report says that “for academic year earnings, a significant difference ($p = 0.038$) was founded between the gender, with men earning more on the average. No difference ($p = 0.476$) was founded between the earnings of black and white students.” Explain both of these conclusions, for the effects of gender and of race on mean earnings, in language understandable to someone who knows no statistics.



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Comparing the Means of Two Related Populations (1 of 4)

- Two approaches that involve related data between groups are possible.
- In the first, items or individuals are **paired** or **matched** according to some characteristics.
- In the second, you take **repeated measurements** from the same set of items or individuals.
- If you know the population standard deviation of the difference scores, use the Z test for the mean difference.



Comparing the Means of Two Related Populations (2 of 4)

Assumptions

- Differences are normally distributed
- Or, if not Normal, use large samples



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Comparing the Means of Two Related Populations (3 of 4)

- The i -th paired difference is D_i , where

$$D_i = X_{1i} - X_{2i}$$

- The point estimate for the paired difference population mean μ_D is $\bar{D}$:

$$\bar{D} = \frac{\sum_{i=1}^n D_i}{n}$$

- The sample standard deviation is S_D :

$$S_D = \sqrt{\frac{\sum_{i=1}^n (D_i - \bar{D})^2}{n - 1}}$$

Note: n is the number of pairs in the paired sample



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Comparing the Means of Two Related Populations (4 of 4)

- Z test for the mean difference

$$Z = \frac{\bar{D} - \mu_D}{\frac{\sigma_D}{\sqrt{n}}}$$

- Paired t test for the mean difference

$$t = \frac{\bar{D} - \mu_D}{\frac{s_D}{\sqrt{n}}}$$

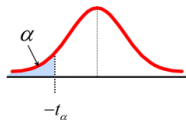


The Paired Difference Test: Possible Hypotheses

Lower-tail test:

$$H_0 : \mu_D \geq 0$$

$$H_1 : \mu_D < 0$$

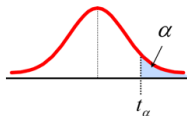


Reject H_0 if $t_{STAT} < -t_\alpha$

Upper-tail test:

$$H_0 : \mu_D \leq 0$$

$$H_1 : \mu_D > 0$$

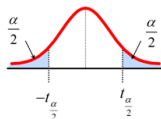


Reject H_0 if $t_{STAT} > t_\alpha$

Two-tail test:

$$H_0 : \mu_D = 0$$

$$H_1 : \mu_D \neq 0$$



Reject H_0 if $t_{STAT} < -t_{\frac{\alpha}{2}}$
or $t_{STAT} > t_{\frac{\alpha}{2}}$



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The Paired Difference Confidence Interval

- The confidence interval for μ_D is:

$$\bar{D} \pm t_{\frac{\alpha}{2}} \frac{S_D}{\sqrt{n}}$$



Example: Paired Difference Test

Assume you send your salespeople to a “customer service” training workshop. Has the training made a difference in the number of complaints? You collect the following data:

Salesperson	Number of Complaints:		(2) - (1) Difference, D_i
	Before (1)	After (2)	
C.B.	6	4	-2
T.F.	20	6	-14
M.H.	3	2	-1
R.K.	0	0	0
M.O.	4	0	<u>-4</u>
			-21



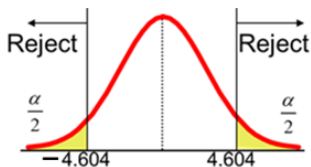
Solution (1 of 3)

- $H_0 : \mu_D = 0$
 $H_1 : \mu_D \neq 0$
- $\alpha = 0.01$, d.f. = $5 - 1 = 4$
- $\bar{D} = \frac{\sum_{i=1}^n D_i}{n} = -4.2$, $S_D = \sqrt{\frac{\sum_{i=1}^n (D_i - \bar{D})^2}{n-1}} = 5.67$
- Critical t values: $t_{0.005} = \pm 4.604$
- Test statistic: $t = \frac{\bar{D} - \mu_D}{\frac{S_D}{\sqrt{n}}} = \frac{-4.2 - 0}{\frac{5.67}{\sqrt{5}}} = \boxed{-1.66}$



Solution (2 of 3)

- Decision: Do not reject H_0
- Conclusion: There is insufficient evidence of a change in the number of complaints.



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Solution (3 of 3)

- The confidence interval for μ_D is: $\bar{D} \pm t_{\frac{\alpha}{2}} \frac{S_D}{\sqrt{n}}$
- $\bar{D} = -4.2$, $S_D = 5.67$
- 99% confidence interval for μ_D :

$$\begin{aligned} & -4.2 \pm 4.604 \frac{5.67}{\sqrt{5}} \\ & = (-15.87, 7.47) \end{aligned}$$

- Since this interval contains 0 you are 99% confident that $\mu_D = 0$.



Comparing Two Population Proportions (1 of 3)

- Goal: Test a hypothesis or form a confidence interval for the difference between two population proportions ($\pi_1 - \pi_2$).
- Assumptions: $n_1\pi_1 \geq 5$, $n_1(1 - \pi_1) \geq 5$
 $n_2\pi_2 \geq 5$, $n_2(1 - \pi_2) \geq 5$
- The point estimate for the difference is $p_1 - p_2$.



Comparing Two Population Proportions (2 of 3)

- In the null hypothesis we assume the null hypothesis is true, so we assume $\pi_1 = \pi_2$ and pool the two sample estimates.
- The pooled estimate for the overall proportion is:

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2}$$

where X_1 and X_2 are the number of items of interest in samples 1 and 2



Comparing Two Population Proportions (3 of 3)

- In evaluating differences between two population proportions, you can use a **Z test for the difference between two proportions**.

$$Z = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1 - \bar{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

$$\text{with } \bar{p} = \frac{X_1 + X_2}{n_1 + n_2} \quad p_1 = \frac{X_1}{n_1} \quad p_2 = \frac{X_2}{n_2}$$



Hypothesis Tests for Two Population Proportions (1 of 2)

- You can use this Z test for the difference between population proportions to determine whether there is a difference in the proportion of successes in the two groups (two-tail test) or whether one group has a higher proportion of successes than the other group (one-tail test).

Two-tail test	One-tail test	One-tail test
$H_0: \pi_1 = \pi_2 \text{ or } \pi_1 - \pi_2 = 0$	$H_0: \pi_1 \geq \pi_2 \text{ or } \pi_1 - \pi_2 \geq 0$	$H_0: \pi_1 \leq \pi_2 \text{ or } \pi_1 - \pi_2 \leq 0$
$H_1: \pi_1 \neq \pi_2 \text{ or } \pi_1 - \pi_2 \neq 0$	$H_1: \pi_1 < \pi_2 \text{ or } \pi_1 - \pi_2 < 0$	$H_1: \pi_1 > \pi_2 \text{ or } \pi_1 - \pi_2 > 0$

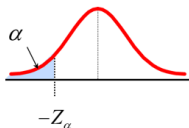


Hypothesis Tests for Two Population Proportions (2 of 2)

Lower-tail test:

$$H_0 : \pi_1 - \pi_2 \geq 0$$

$$H_1 : \pi_1 - \pi_2 < 0$$

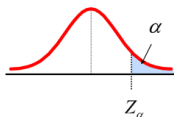


Reject H_0 if $Z_{STAT} < -Z_\alpha$

Upper-tail test:

$$H_0 : \pi_1 - \pi_2 \leq 0$$

$$H_1 : \pi_1 - \pi_2 > 0$$

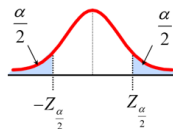


Reject H_0 if $Z_{STAT} > Z_\alpha$

Two-tail test:

$$H_0 : \pi_1 - \pi_2 = 0$$

$$H_1 : \pi_1 - \pi_2 \neq 0$$



Reject H_0 if $Z_{STAT} < -Z_{\frac{\alpha}{2}}$
or $Z_{STAT} > Z_{\frac{\alpha}{2}}$



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Example: Two Population Proportions

Is there a significant difference between the proportion of men and the proportion of women who will vote Yes on Proposition A?

- In a random sample, 36 of 72 men and 35 of 50 women indicated they would vote Yes.
- Test at the 0.05 level of significance.



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Solution (1 of 3)

- The hypothesis test is:
 $H_0 : \pi_1 - \pi_2 = 0$ (the two proportions are equal)
 $H_1 : \pi_1 - \pi_2 \neq 0$ (there is a significant difference between proportions)
- The sample proportions are:
 - Men: $p_1 = \frac{36}{72} = 0.50$
 - Women: $p_2 = \frac{35}{50} = 0.70$
- The pooled estimate for the overall proportion is:

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{36 + 35}{72 + 50} = \frac{71}{122} = 0.582$$



Solution (2 of 3)

- The test statistic for $\pi_1 - \pi_2$ is:

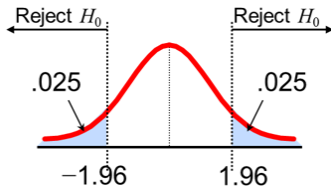
$$\begin{aligned} Z &= \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1 - \bar{p})(\frac{1}{n_1} + \frac{1}{n_2})}} \\ &= \frac{(0.50 - 0.70) - (0)}{\sqrt{0.582(1 - 0.582)(\frac{1}{72} + \frac{1}{50})}} = \boxed{-2.20} \end{aligned}$$

- Critical Z values = ± 1.96 (for $\alpha = 0.05$)



Solution (3 of 3)

- Decision: Reject H_0
- Conclusion: There is evidence of a significant difference in the proportion of men and women who will vote yes.



Confidence Interval for Two Population Proportions

- The confidence interval for $\pi_1 - \pi_2$ is:

$$(p_1 - p_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{p_1(1 - p_1)}{n_1} + \frac{p_2(1 - p_2)}{n_2}}$$

- The 95% confidence interval for $\pi_1 - \pi_2$ is

$$\begin{aligned} (0.50 - 0.70) \pm 1.96 \sqrt{\frac{0.50(0.50)}{72} + \frac{0.70(0.30)}{50}} \\ = (-0.37, -0.03) \end{aligned}$$

- Since this interval does not contain 0, we can be 95% confident the two proportions are different.



Testing for the Ratio of Two Population Variances

Hypotheses	F_{STAT}
$H_0 : \sigma_1^2 = \sigma_2^2$	
$H_1 : \sigma_1^2 \neq \sigma_2^2$	$\frac{S_1^2}{S_2^2}$
$H_0 : \sigma_1^2 \leq \sigma_2^2$	
$H_1 : \sigma_1^2 > \sigma_2^2$	

Where:

S_1^2 = Variance of sample 1 (the larger sample variance)

n_1 = sample size of sample 1

S_2^2 = Variance of sample 2 (the smaller sample variance)

n_2 = sample size of sample 2

$n_1 - 1$ = numerator degrees of freedom

$n_2 - 1$ = denominator degrees of freedom



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The F Distribution

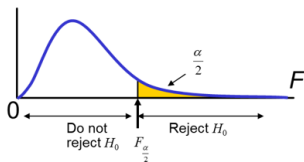
- There are two degrees of freedom required: numerator and denominator.
- The larger sample variance is always the numerator.
- When $F = \frac{S_1^2}{S_2^2}$, $df_1 = n_1 - 1$, $df_2 = n_2 - 1$



Finding the Rejection Region

$$H_0 : \sigma_1^2 = \sigma_2^2$$

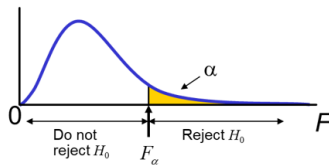
$$H_1 : \sigma_1^2 \neq \sigma_2^2$$



Reject H_0 if $F_{STAT} > F_{\frac{\alpha}{2}}$

$$H_0 : \sigma_1^2 \leq \sigma_2^2$$

$$H_1 : \sigma_1^2 > \sigma_2^2$$



Reject H_0 if $F_{STAT} > F_{\alpha}$



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Finding Lower Tail Critical Values

- You calculate F_L , a lower-tail critical value on the F distribution with $n_1 - 1$ degrees of freedom in the numerator and $n_2 - 1$ degrees of freedom in the denominator, by taking the reciprocal of F_U^* , an upper-tail critical value on the F distribution with degrees of freedom 'switched' (i.e. $n_2 - 1$ degrees of freedom in the numerator and $n_1 - 1$ degrees of freedom in the denominator).

$$F_L = \frac{1}{F_U^*}$$



Example: F Test

You are a financial analyst for a brokerage firm. You want to compare dividend yields between stocks listed on the NYSE & NASDAQ. You collect the following data:

	NYSE	NASDAQ
Number	21	25
Sample mean	3.27	2.53
Sample std dev	1.30	1.16

Is there a difference in the variances between the NYSE & NASDAQ at the $\alpha = 0.05$ level?



Solution (1 of 2)

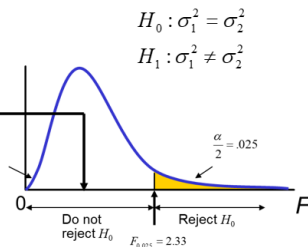
- Form the hypothesis test:
 $H_0 : \sigma_1^2 = \sigma_2^2$ (there is no difference between variances)
 $H_1 : \sigma_1^2 \neq \sigma_2^2$ (there is a difference between variances)
- Find the F critical value for $\alpha = 0.05$.
- Numerator d.f. = $n_1 - 1 = 21 - 1 = 20$
- Denominator d.f. = $n_2 - 1 = 25 - 1 = 24$
- $F_{\frac{\alpha}{2}} = F_{0.975, 20, 24} = 2.33$



Solution (2 of 2)

- The F test statistic is:

$$F_{STAT} = \frac{S_1^2}{S_2^2} = \frac{1.30^2}{1.16^2} = \boxed{1.256}$$



- Decision: $F = 1.256$ is not in the rejection region, so we do not reject H_0
- Conclusion: There is not sufficient evidence of a difference in variances at $\alpha = 0.05$.



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